THARUN G

AI / Machine Learning Engineer

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SUMMARY

Third-year CSE student building applied AI systems: LSTM-based anomaly detection with explainability, agentic workflows on local LLMs, and hybrid RAG memory. AWS AI & ML Scholar with a spatial-ML internship. Seeking an AI/ML engineering internship or entry-level role.

EDUCATION

B.E. Computer Science and Engineering | National Engineering College, Kovilpatti 2024 – 2028 (Expected)
CGPA: 6.8 / 10

SKILLS

AI/ML: PyTorch, scikit-learn, Deep Learning, LSTMs, Computer Vision, Generative AI, LangGraph, Ollama, RAG (FAISS, BM25), Random Forest, Data Preprocessing
Engineering: Python, FastAPI, Rust (PyO3), SQL, Docker, AWS, Model Deployment, Git
Data & Visualization: Pandas, NumPy, Streamlit, Plotly, GeoPandas, PySAL

PROJECTS

Astrovia: Explainable Satellite Anomaly Detection | Python, PyTorch, scikit-learn, SatNOGS, Streamlit, Plotly, Docker SIH 2026 (Waitlisted)

- Replaced fixed-threshold telemetry alarms with per-channel PyTorch LSTM forecasters and z-score scoring of prediction residuals; validated on NASA SMAP and MSL (Curiosity) datasets.
- Built an explainability layer based on prediction-error attribution that ranks root-cause sensors without the compute cost of SHAP.
- Ingested live satellite frames through SatNOGS decoders and built a Streamlit/Plotly mission-control dashboard with severity scoring and root-cause views; containerized with Docker.

Orion AI: Local-First Personal AI Assistant | Python, Rust (PyO3), FastAPI, Ollama, FAISS, SQLite FTS5 github.com/AstraDev-Labs/Orion-AI

- Built a fully offline assistant on a five-part design (intelligence, engine, agentic logic, memory, learning) running local quantized models through Ollama.
- Implemented hybrid RAG memory combining SQLite FTS5/BM25 and FAISS dense retrieval with Reciprocal Rank Fusion.
- Built ReAct and CodeAct agent loops with loop guards, Docker/Wasm sandboxed execution, human approval for sensitive actions and a trace store for evaluating agent runs.
- Moved performance-critical components (scanning, prompt-injection detection, vector retrieval) to Rust via PyO3 and shipped a Tauri 2 + React desktop app with local speech input and output.

SentinelAI: Autonomous SIEM & SOAR Assistant | Python, LangGraph, Ollama (Llama 3.1), Redpanda, OpenSearch, Docker

- Built a LangGraph triage agent running a local Llama 3.1 model with deterministic fallbacks; it scores security alerts and triggers allow-listed containment playbooks.
- Built the supporting streaming pipeline (Vector, Redpanda, OpenSearch) and compiled Sigma rules into queries for MITRE ATT&CK detections.

EXPERIENCE

Spatial Data Engineering Intern | Thazhal Geospatial June 2026

- Built a full-stack spatial AI tool (Next.js, Python, GeoPandas) that estimates Total Addressable Market for financial-infrastructure expansion.
- Wrote offline pipelines to ingest, clean and spatially aggregate 23,000+ commercial locations from the Google Places API into municipal boundaries.
- Applied PySAL spatial lag and Random Forest models to account for competitor density when scoring candidate locations.

CERTIFICATIONS

- AWS AI & ML Scholar and AWS AI Practitioner Challenge (Udacity, 2026)
- Discovering Cybersecurity (Udacity & Accenture, 2026)
- NPTEL: The Joy of Computing using Python, IIT Madras (69%)